

Appendix

Missing Level	Feature	Method	RMSE	MAE	Wilcoxon Stat	Wilcoxon P-Value
5	Gaze X	mean	376.61	284.35	3995126.00	0.98
5	Gaze X	locf	447.01	332.72	3597955.50	0.00
5	Gaze X	knn	379.46	286.20	3987326.50	0.89
5	Gaze X	gpr	491.76	367.71	2567092.00	0.00
5	Gaze X	mice	391.32	309.00	3440535.00	0.00
5	Gaze Y	mean	261.88	159.88	925211.50	0.00
5	Gaze Y	locf	300.08	205.63	703906.00	0.00
5	Gaze Y	knn	274.99	180.95	1184036.00	0.35
5	Gaze Y	gpr	370.01	282.96	532362.00	0.00
5	Gaze Y	mice	262.03	163.49	1093712.50	0.00
5	ET_PupilLeft	mean	2.13	0.96	4543330.50	0.00
5	ET_PupilLeft	locf	2.10	0.88	4707305.00	0.23
5	ET_PupilLeft	knn	2.12	0.92	4704685.00	0.05
5	ET_PupilLeft	gpr	2.12	0.90	4726783.50	0.01
5	ET_PupilLeft	mice	2.12	0.92	4765974.50	0.03
5	ET_PupilRight	mean	2.03	0.94	4590584.50	0.00
5	ET_PupilRight	locf	1.99	0.85	4706863.50	0.32
5	ET_PupilRight	knn	2.02	0.89	4687577.50	0.03
5	ET_PupilRight	gpr	2.02	0.88	4748517.50	0.01
5	ET_PupilRight	mice	2.02	0.89	4766748.50	0.02
10	Gaze X	mean	374.59	282.43	3940903.00	0.44
10	Gaze X	locf	449.85	334.10	3595862.00	0.00
10	Gaze X	knn	380.70	287.90	3901035.50	0.19
10	Gaze X	gpr	493.19	368.95	2553563.00	0.00
10	Gaze X	mice	387.59	304.87	3452881.00	0.00
10	Gaze Y	mean	257.99	159.26	929634.00	0.00
10	Gaze Y	locf	299.17	205.45	701674.00	0.00
10	Gaze Y	knn	265.00	172.14	1155402.50	0.06
10	Gaze Y	gpr	370.28	283.49	527007.50	0.00
10	Gaze Y	mice	258.55	163.77	1106565.00	0.00
10	ET_PupilLeft	mean	2.14	1.02	4477045.50	0.00
10	ET_PupilLeft	locf	2.10	0.88	4714372.50	0.25
10	ET_PupilLeft	knn	2.11	0.93	5052402.50	0.84
10	ET_PupilLeft	gpr	2.12	0.92	4745085.50	0.00
10	ET_PupilLeft	mice	2.13	0.93	4856077.50	0.01
10	ET_PupilRight	mean	2.04	0.99	4564128.50	0.00
10	ET_PupilRight	locf	2.00	0.85	4707896.00	0.29
10	ET_PupilRight	knn	2.01	0.89	4696188.00	0.03
10	ET_PupilRight	gpr	2.02	0.89	4790717.50	0.00
10	ET_PupilRight	mice	2.01	0.89	4824425.50	0.00

Table 7. Table with RMSE, MAE and Wilcoxon-Test for different methods

Note: Percentage drops are computed relative to the pre-normalization variance. Very small denominators can yield seemingly extreme percentages; interpretation should focus on qualitative magnitude (“near-complete”) rather than absolute percent.

Missing Level	Feature	Method	RMSE	MAE	Wilcoxon Stat	Wilcoxon P-Value
15	Gaze X	mean	366.75	277.75	3951423.00	0.51
15	Gaze X	locf	448.60	333.15	3600427.00	0.00
15	Gaze X	knn	385.20	282.96	3578735.00	0.00
15	Gaze X	gpr	492.16	368.51	2536686.00	0.00
15	Gaze X	mice	409.68	324.86	3145983.00	0.00
15	Gaze Y	mean	256.06	156.55	985295.50	0.00
15	Gaze Y	locf	297.68	203.39	709170.00	0.00
15	Gaze Y	knn	258.79	165.21	1201433.50	0.62
15	Gaze Y	gpr	368.75	282.29	527888.00	0.00
15	Gaze Y	mice	257.06	165.21	1208120.50	0.76
15	ET_PupilLeft	mean	2.16	1.09	4475079.00	0.00
15	ET_PupilLeft	locf	2.09	0.87	4664690.50	0.31
15	ET_PupilLeft	knn	2.12	0.93	4763183.00	0.06
15	ET_PupilLeft	gpr	2.13	0.93	4809430.00	0.00
15	ET_PupilLeft	mice	2.12	0.94	5012904.00	0.03
15	ET_PupilRight	mean	2.05	1.05	4406991.50	0.00
15	ET_PupilRight	locf	1.98	0.84	4714432.50	0.34
15	ET_PupilRight	knn	2.01	0.89	4735521.00	0.03
15	ET_PupilRight	gpr	2.02	0.90	4737550.50	0.00
15	ET_PupilRight	mice	2.01	0.90	4865398.50	0.00
20	Gaze X	mean	363.54	275.61	3963029.50	0.62
20	Gaze X	locf	446.13	330.48	3497413.00	0.00
20	Gaze X	knn	376.95	282.78	3826664.00	0.02
20	Gaze X	gpr	554.23	433.91	1699713.50	0.00
20	Gaze X	mice	383.68	303.59	3411282.50	0.00
20	Gaze Y	mean	257.74	159.40	1042185.50	0.00
20	Gaze Y	locf	301.81	210.35	660598.00	0.00
20	Gaze Y	knn	264.68	172.34	1207094.00	0.68
20	Gaze Y	gpr	372.13	285.44	522901.00	0.00
20	Gaze Y	mice	260.94	170.80	1168706.50	0.08
20	ET_PupilLeft	mean	2.17	1.14	4395822.50	0.00
20	ET_PupilLeft	locf	2.10	0.88	4648368.50	0.21
20	ET_PupilLeft	knn	2.14	0.95	4785724.00	0.06
20	ET_PupilLeft	gpr	2.16	0.96	4699989.50	0.00
20	ET_PupilLeft	mice	2.14	0.97	5050027.50	0.00
20	ET_PupilRight	mean	2.08	1.11	4420682.50	0.00
20	ET_PupilRight	locf	1.99	0.85	4730801.50	0.38
20	ET_PupilRight	knn	2.03	0.92	4685272.00	0.00
20	ET_PupilRight	gpr	2.03	0.93	4775423.50	0.00
20	ET_PupilRight	mice	2.03	0.93	4799654.50	0.00

Table 8. Table with RMSE, MAE and Wilcoxon-Test for different methods

Outlier Level	Feature	Method	Variance Reduction (%)	KS-Test Statistic	KS-Test P-Value
5	Gaze X	Z-Score	0.00	0.00	1.00
5	Gaze Y	Z-Score	0.00	0.00	1.00
5	ET_PupilLeft	Z-Score	0.00	0.000185	1.00
5	ET_PupilRight	Z-Score	0.00	0.00	1.00
5	Gaze X	MAD	442000	0.117	5.78×10^{-23}
5	Gaze Y	MAD	147000	0.107	8.64×10^{-11}
5	ET_PupilLeft	MAD	32.0	0.245	1.35×10^{-117}
5	ET_PupilRight	MAD	12.2	0.250	3.69×10^{-122}
5	Gaze X	Isolation Forest	382000	0.049	1.62×10^{-4}
5	Gaze Y	Isolation Forest	118000	0.047	0.0154
5	ET_PupilLeft	Isolation Forest	26.1	0.049	4.02×10^{-6}
5	ET_PupilRight	Isolation Forest	7.30	0.049	3.99×10^{-6}
10	Gaze X	Z-Score	0.00	0.00	1.00
10	Gaze Y	Z-Score	0.00	0.00	1.00
10	ET_PupilLeft	Z-Score	0.00	0.000185	1.00
10	ET_PupilRight	Z-Score	0.00	0.00	1.00
10	Gaze X	MAD	577000	0.159	4.57×10^{-41}
10	Gaze Y	MAD	346000	0.148	5.88×10^{-20}
10	ET_PupilLeft	MAD	34.6	0.231	1.61×10^{-101}
10	ET_PupilRight	MAD	17.1	0.239	2.84×10^{-108}
10	Gaze X	Isolation Forest	208000	0.040	0.00328
10	Gaze Y	Isolation Forest	109000	0.043	0.0383
10	ET_PupilLeft	Isolation Forest	10.6	0.047	1.70×10^{-5}
10	ET_PupilRight	Isolation Forest	4.37	0.047	1.65×10^{-5}

Table 9. Outlier Detection Evaluation

Outlier Level	Feature	Method	Variance Reduction (%)	KS-Test Statistic	KS-Test P-Value
15	Gaze X	Z-Score	0.00	0.00	1.00
15	Gaze Y	Z-Score	0.00	0.000903	1.00
15	ET_PupilLeft	Z-Score	0.00	0.000185	1.00
15	ET_PupilRight	Z-Score	0.00	0.000185	1.00
15	Gaze X	MAD	482000	0.173	1.04×10^{-47}
15	Gaze Y	MAD	269000	0.200	8.77×10^{-35}
15	ET_PupilLeft	MAD	57.8	0.219	6.69×10^{-88}
15	ET_PupilRight	MAD	31.7	0.223	6.00×10^{-91}
15	Gaze X	Isolation Forest	140000	0.035	0.0150
15	Gaze Y	Isolation Forest	53600	0.039	0.0767
15	ET_PupilLeft	Isolation Forest	4.33	0.012	0.762
15	ET_PupilRight	Isolation Forest	3.84	0.045	5.06×10^{-5}
20	Gaze X	Z-Score	0.00	0.00	1.00
20	Gaze Y	Z-Score	0.00	0.00181	1.00
20	ET_PupilLeft	Z-Score	0.00	0.00	1.00
20	ET_PupilRight	Z-Score	0.00	0.000185	1.00
20	Gaze X	MAD	1460000	0.199	3.91×10^{-62}
20	Gaze Y	MAD	575000	0.214	1.00×10^{-39}
20	ET_PupilLeft	MAD	78.0	0.205	3.84×10^{-74}
20	ET_PupilRight	MAD	44.4	0.210	1.17×10^{-77}
20	Gaze X	Isolation Forest	222000	0.034	0.0188
20	Gaze Y	Isolation Forest	76200	0.038	0.0837
20	ET_PupilLeft	Isolation Forest	5.51	0.017	0.395
20	ET_PupilRight	Isolation Forest	3.04	0.016	0.433

Table 10. Outlier Detection Evaluation

MV	Outlier	Normalization	Feature	Variance Red. (%)	Skewness Red.	Kurtosis Red.	KS-Test
knn	mad	Min-Max	Gaze X	100.00	0.00	0.00	1
			Gaze Y	100.00	2.33E-15	4.44E-16	1
			ET_PupilLeft	48.68	3.88E-15	0.00	1
	zscore	zscore	ET_PupilRight	29.91	2.38E-15	0.00	1
			Gaze X	99.99	0.00	0.00	1
			Gaze Y	99.98	8.88E-16	0.00	1
locf	mad	Min-Max	ET_PupilLeft	-1657.86	5.44E-15	0.00	1
			ET_PupilRight	-1847.27	1.88E-15	0.00	1
	zscore	zscore	Gaze X	100.00	1.11E-15	0.00	1
			Gaze Y	100.00	0.00	0.00	1
			ET_PupilLeft	43.18	1.78E-15	0.00	1
	zscore	zscore	ET_PupilRight	25.47	0.00	1.33E-15	1
isolationforest	mad	Min-Max	Gaze X	100.00	3.05E-16	0.00	1
			Gaze Y	100.00	0.00	0.00	1
			ET_PupilLeft	-1672.13	1.67E-15	0.00	1
	zscore	zscore	ET_PupilRight	-1856.96	0.00	8.88E-16	1
			Gaze X	100.00	4.44E-16	8.88E-16	1
			Gaze Y	100.00	0.00	0.00	1
isolationforest	Min-Max	Min-Max	ET_PupilLeft	97.15	0.00	1.33E-15	0.7337
			ET_PupilRight	96.65	2.22E-16	0.00	0.7289
	zscore	zscore	Gaze X	100.00	0.00	0.00	1
			Gaze Y	99.99	0.00	4.44E-16	1
	Min-Max	Min-Max	ET_PupilLeft	82.69	2.22E-16	4.44E-16	0.7337
			ET_PupilRight	80.00	2.22E-16	0.00	0.7289

Table 11. Variance, Skewness, and Kurtosis Reduction with KS-Test Statistics